



education
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GAUTENG PROVINCE

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ANNUAL TEACHING PLAN NATURAL SCIENCE GRADE 5- YEAR 1-3

ATP Grade 5 : Term 1

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1	Plants, animals, birds and insects on earth	• Know and understand the following terminology: habitats, indigenous plants • Identify and name the habitats (home) of different plants e.g. weeds, water lily, aloe, protea bush, fruit tree • Identify the colour and/or shape of the leaves/flowers/fruit • Identify and name the habitats of different animals, birds and insects e.g. impala, birds and spiders • Know and understand that plants, animals, birds and insects can have	The teacher takes the learners to the school garden and shows them the different types of habitats and plants, animals, birds and insects <u>Grade 5: First year</u> <u>Activity 1</u> Learners work together in pairs. Count the number of different plants in a small section of the garden that you have chosen. Write the names of the plants that you know and the number of plants in your book. <u>Activity 2</u> Study a picture and list the living and non-living things in the picture. Name three non-living things that the living things depend on. <u>Grade 5: Second year</u> <u>Activity 1</u>

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		more than one habitat - Explain why plants, animals, birds and insects depend on each other for food and protection - State and explain that plants also depend on plants, animals, birds and insects for food - Explain that animals need non-living things such as water, air and sunlight to grow	Learners work together in pairs. Count the number of different plants, animals, birds and insects in a small section of the garden that you have chosen. Write the names of the plants that you know and the number of plants in your book. <u>Activity 2</u> Group the plants in groups according to their habitat, as either plants living in water or soil. <u>Grade 5: Third year</u> <u>Activity 1</u> Draw a table and record the plants, animals, birds and insects according to the habitat, number, shape and colour. <u>Activity 2</u> Collect a leave and flower (if available) from the plants, dry between newspapers, paste into your book and label correctly.
2	Plants animals, birds and insects on earth	- Know how plants, animals, birds and insects reproduce, namely with seeds/cuttings (plants), live births (animals and humans) and eggs (birds and insects) - Know that there are many different kinds of animals, birds and insects - Identify and name animals, birds and insects with bone and without bones (e.g. spiders, crabs, beetles, crayfish, cockroach, crickets, grasshoppers)	The teacher shows pictures/models of animals, birds and insects to enable learners to complete the activities. <u>Grade 5: First year</u> <u>Activity 1</u> Draw a table with two columns and group the animals, birds and insects in the pictures in two groups, namely animals, birds and insects without bones and with bones. <u>Grade 5 : Second year</u> <u>Activity 1</u> Draw a table with two columns and group the animals, birds and insects in the pictures in two groups, namely without bones and with bones.

Grade 5 : Term 1

Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Distinguish animals without bones by their hard outer shell from animals with bones - State the function of a hard outer shell - State the function of a skeleton, namely to support the body	<u>Activity 2</u> List two animals and insects with a hard outer shell and give the reason why they have this hard outer shell. <u>Grade 5: Third year</u> <u>Activity 1</u> Draw a table with two columns and group the animals, birds and insects in the pictures in two groups, namely without bones and with bones. <u>Activity 2</u> List how the animals, birds and insects reproduce. <u>Activity 3</u> List two animals with skeletons and give the reasons why they have skeletons. <u>Activity 4</u> Explain the word: “dependent”.
3-4	Animal skeletons	- Know and understand the terminology: skeleton, joints, muscles - Identify the basic structure of animals, birds and insects namely head, tail, body, limbs, sense organs and the visible differences between animals, birds and insects such as size, shape, body covering and sense organs - Explain that a skeleton of animals consists of bones and joints and is	The teacher does an experiment with two chicken bones. The one chicken bone is placed in a cup filled with water and the other in a cup filled with vinegar to demonstrate what happens when bones do not get calcium. The learners discuss the outcome of the experiment. Learners - do different exercises in the class and feel how their muscles and friends’ muscles contract - feel their spine and ribs - count their ribs - bend and straighten your elbow to identify a hinge joint

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		inside the body - List the main skeleton parts of animals, namely skull, backbone, ribs, arms, legs and hip bones - List the functions of each animal skeleton part, namely to protect the brain, protect the spinal cord, protect the lungs and heart and the arms, legs and hip bones provide for movement - State that calcium in bones make them strong - State the function of muscles, namely to move the bones in the body - Explain that bones move when muscles contract and feel hard - Identify and name joints in the body as the places where bones meet - Identify how animals, birds and insects move, e.g. walking, swimming, hopping, flying, trotting - Give examples of shell structures, namely crabs, snails and oysters	- swing your arm at the shoulder to identify a ball-and-socket joint <u>Grade 5: First year</u> <u>Activity 1</u> Label the skeleton parts in the picture of a fish. <u>Activity 2</u> Lift a weight and let your friend feel how your arm muscles contract. Explain how bones in the body move. <u>Activity 3</u> Work with a friend and gather material to make a fish skeleton according to the image provided to you. You will continue with this activity next week. <u>Grade 5: Second year</u> <u>Activity 1</u> Label the skeleton parts in the picture of a bird. Explain how the skeleton parts of a fish differ from a fish's. <u>Activity 2</u> Explain why we are able to bend our arms and legs and other body parts. <u>Activity 3</u> Work with a friend and gather material to make a bird skeleton according to the image provided to you. You will continue with this activity next week. <u>Grade 5: Third year</u> <u>Activity 1</u>

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			Compare the skeleton parts of a human, fish and bird. How do they differ? <u>Activity 2</u> Label the skeleton parts and joints in the human body on the picture. <u>Activity 3</u> Work with a friend and gather material to make a human skeleton according to the image provided to you. You will continue with this activity next week.
5	Food chains	• Tell why living things need food and where they get food from • Discuss how plants make their own food • Explain why animals need food • Identify the food animals eat, namely plants only, meat only or both plants and animals • List examples of animals that eat plants (e.g. cattle and sheep), meat (lions) and both plants and meat (dogs and pigs) • Describe a food chain, namely that it starts with a plant, then follows with an animal that eats the plant, then an animal that eats an animal and the human that eats the animal • Sequence pictures of different food	The teacher shows - images of different types of animals and group them together according to what they eat, namely plants only, meat only or both plants and animals - a video of short and longer food chains The teacher and learners identify living creatures in the school grounds and without hurting them, study their mouths with a magnifying glass, e.g. spiders, snails, ants, birds, lizards, chickens or caterpillars. <u>Grade 5: First year</u> <u>Activity 1</u> Label the picture of how plants make their own food <u>Activity 2</u> Make a two dimensional model illustrating the food chain depicting 3 living things. <u>Activity 3</u> Select one animal found on the school grounds and find a picture in a magazine of the animal. Name the animal, the type of food the animal eats and where the animal was found on the school grounds.

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		chains in the correct sequence Know and understand the terminology: herbivores, carnivores and omnivores	<u>Grade 5: Second year</u> <u>Activity 1</u> Copy the teacher's example of how plants make their own food. <u>Activity 2</u> Make a three dimensional model illustrating the food chain depicting 3 living things. <u>Grade 5: Third year</u> <u>Activity 1</u> Draw a picture of how plants make their own food <u>Activity 2</u> Select and group images of animals that eat plants, meat and both plants and meat. Name each animal in the image. <u>Activity 3</u> Make a three dimensional model illustrating the food chain depicting 4 living things.
6	Life cycles	- Understand the terminology: "life cycle" - Describe the life cycle of a human - Describe the life cycle of a plant - Know and understand the terminology: pollinate, germinate, fertilisation - Describe the life cycle of an animal - Know and understand the terminology: pregnancy, mammals, fertilisation, reproduce	<u>Grade 5: First year</u> <u>Activity 1</u> Cut a flower and dissect the flower. Identify the different parts of the flower, namely petals, style, anther, stigma and ovary. Complete the worksheet by labelling the parts of the flower. <u>Activity 2</u> Arrange the pictures showing the life cycle of a mammal in the correct sequence and paste in your workbook. <u>Grade 5: Second year</u>

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Describe the life cycle of a bird - Describe the life cycle of an insect	<u>Activity 1</u> Cut a flower and dissect the flower. Identify the different parts of the flower, namely petals, style, anther, stigma and ovary. Copy the image of a flower and label the parts of the flower. <u>Activity 2</u> Arrange the pictures showing the life cycle of a bird in the correct sequence and paste in your workbook. <u>Grade 5: Third year</u> <u>Activity 1</u> Describe the following words: life cycle, pollinate, germinate, fertilisation. <u>Activity 2</u> Look at the pictures of the life cycle of a mammal and describe each picture. <u>Activity 3</u> Dissect an insect and label the parts of the insect. <u>Activity 4</u> Describe the life cycle of a moth or fly.
7	Nutrients in food	- Label the SA food pyramid correctly - Know and understand the terminology nutrients - List nutrients in the bread and pasta group as starch, meat, fish and poultry group as protein and fat, dairy group as protein, calcium and water, and the fruit as well as vegetable group as vitamins	The teacher - demonstrates how to test food for starch with iodine. Use a starch rich food, e.g. a slice of bread and a non-starchy food as a control experiment - shows learners how to examine food for fat content, by putting a tissue or paper towel over the food as the fat/oiliness will be absorbed by the tissue or paper towel. - demonstrates an experiment to test for sugar in food using Benedict's solution (videos of experiments available on: http://www.bbc.co.uk/bitesize/ks3/science/organisms_behaviour_health/food_detective

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		and water - Describe how to prepare and eat fruit and vegetables to ensure that you do not waste nutrients - Explain how to store bread, pasta, fruit, vegetable, meat and dairy products to make sure that the food stays fresh and not lose their nutritional value	/activity/ <u>Grade 5: First year</u> <u>Activity 1</u> List the nutrients that we find in food, namely starch, protein, calcium, vitamins and water. <u>Activity 2</u> Execute the experiment to test food for starch. <u>Grade 5: Second year</u> <u>Activity 1</u> List one food source containing each of the following nutrients: starch, protein, calcium, vitamins and water. <u>Activity 2</u> Perform the experiment to test food for fat content. <u>Grade 5: Third year</u> <u>Activity 1</u> Identify the nutrient in each of the following food sources: milk, biltong, fish, white bread, apples, carrots, yoghurt, cheese, bananas, and spinach. <u>Activity 2</u> Perform the experiment to test food for sugar.
8	Digestion	- Explain that the food we eat is digested - Tell that teeth and saliva help to digest food - Explain that organs in the body break	<u>Grade 5: First year</u> <u>Activity 1</u> Identify the organs on the image that digest food. <u>Activity 2</u>

Grade 5 : Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		food down/digest food and absorb the nutrients from the food - State that undigested material is waste material and the body gets rid of the waste through the rectum - Know and understand the terminology: digestion, digestive system, saliva	Place three digestive organs in the correct place in a model. <u>Grade 5: Second year</u> <u>Activity 1</u> Order the organs in the image in the correct order of how they assist with digestion. <u>Activity 2</u> Place four digestive organs in the correct place in a model. <u>Grade 5: Third year</u> <u>Activity 1</u> Label the organs in the image of the digestive system correctly. <u>Activity 2</u> Place all the digestive organs in the correct place in a model.
9	Recycling	- Know how to group bottles, cans, paper, electronic waste and plastic correctly - Know how to re-use recyclables	<u>Grade 5: First year</u> <u>Activity 1</u> Group recyclables and practise to recycle the items. <u>Grade 5: Second year</u> <u>Activity 1</u> Group recyclables and practise to recycle the items. <u>Grade 5: Third year</u> <u>Activity 1</u> Group recyclables and practise to recycle the items.



2.1. Grade 5 Term 2

Strand: Matter and materials

Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1 - 2	Metals and non-metals	• Arrange objects in groups of metal and non-metal objects • List the properties of metals, namely that metals are strong, very hard, shiny, malleable, conducts heat, is magnetic and melt at high temperatures • List objects made with iron, namely tools, nails, fence posts, wire fencing, palisade fencing • Know and describe the terminology: rust, alloy • List objects made with gold, silver copper and aluminium • Explain why articles are plated • List examples of non-metals • List the properties of non-metals, namely that they are not malleable and not ductile • Know and understand the terminology: rust, alloy, plated, non-metals, malleable	The teacher does an experiment with a plastic and metal spoon to show the hardness of metal. - Place the plastic and metal spoon side by side on a table with part of the spoons hanging over the side of the table. Gently push the spoons down to show that metal is hard and do not flexible. Explain the difference between hardness, flexibility and malleable. spoons - Stand on the metal and plastic spoon to demonstrate hardness of the metal spoon <u>Grade 5: First year</u> <u>Activity 1</u> Bring five items from home and identify what it is used for. <u>Activity 2</u> Group the items as metals or non-metals. <u>Grade 5: Second year</u> <u>Activity 1</u> Bring five pictures from home and identify what it is used for. <u>Activity 2</u> Group the items as metals or non-metals and list the properties of metals. <u>Grade 5: Third year</u> <u>Activity 1</u>

Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			Make a list of five metals and write a sentence to describe what these are used for. <u>Activity 2</u> Paste five pictures of non-metals in your book and list the properties of the non-metals.
3-4	Uses of metals	• Demonstrate three properties of metals, namely that metals rust, conduct heat and show magnetic properties • Demonstrate the opposite sides of a bar magnet attract each other • Know and understand the terminology: tarnish, corrode and magnetism	The teacher does an experiment to show learners that - metals rust - conducts heat (put butter on the tips of metal and non-metal objects and place other end in boiling water) - exhibits magnetic properties (e.g. use iron filings) - the opposite poles of magnets attract each other (e.g. kissing fish experiment) <u>Grade 5: First year</u> <u>Activity 1</u> Perform an experiment to show that metals rust and use two different articles in your experiment. Record your findings in your workbooks. <u>Activity 2</u> Perform an experiment to show that metals conduct heat. Record the objects that you used to execute the experiment and findings in your workbooks. <u>Grade 5: Second year</u> <u>Activity 1</u> Perform an experiment to show that metals rust and use four different articles in your experiment. Record your findings in your workbooks. <u>Activity 2</u> Perform an experiment to show that the opposite poles of a bar magnet attract each other. Record the objects that you used to execute the experiment and findings in your workbooks.

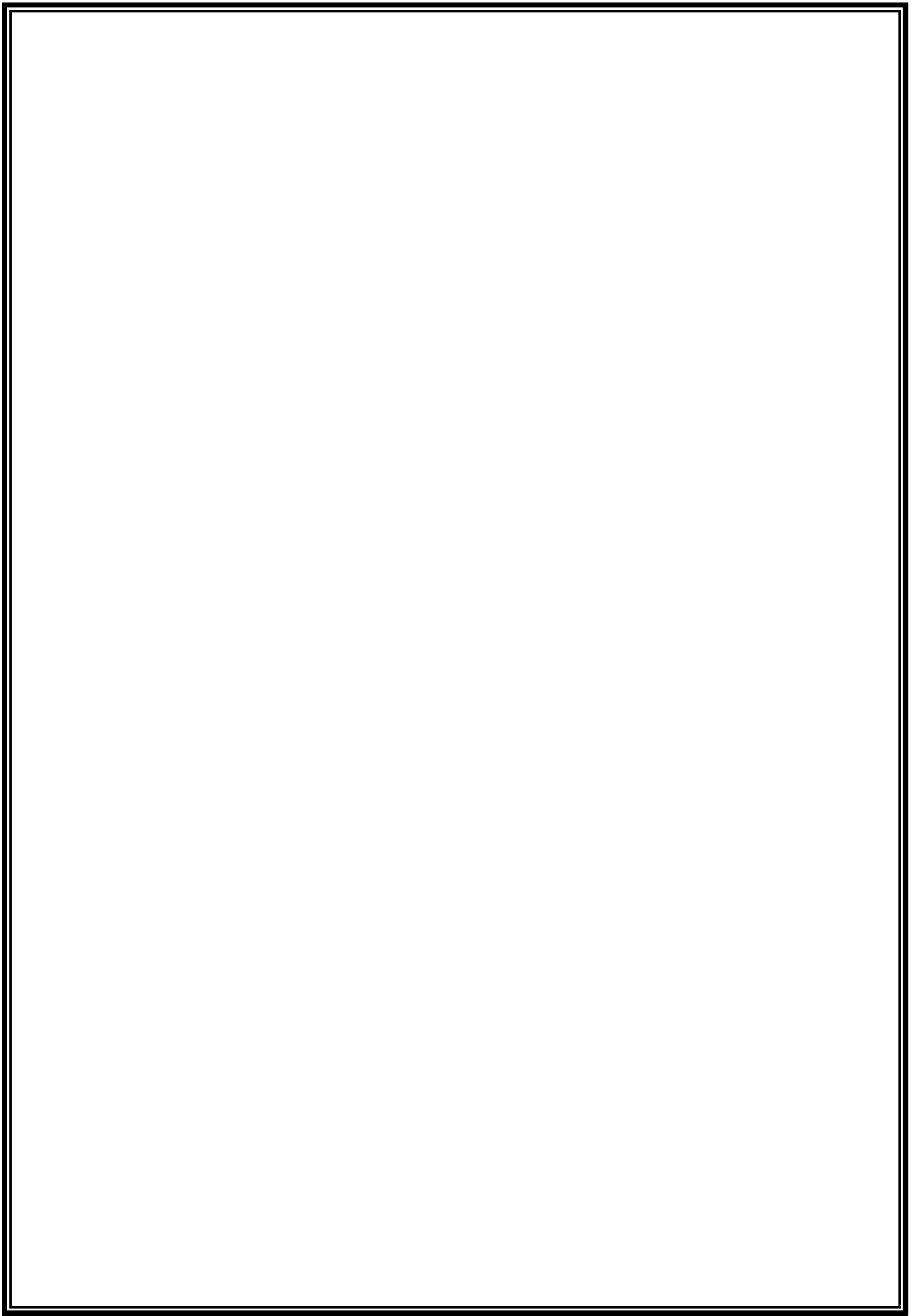
Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			<u>Grade 5: Third year</u> <u>Activity 1</u> Perform an experiment to show that metals rust and non-metals do not rust. Use the six articles provided to you in your experiment. Record your findings in your workbooks. <u>Activity 2</u> Perform an experiment to show that metals are magnetic by using a bar magnet and iron filings. Record one of the patterns made with the iron filings in your workbooks.
5-6	Processing materials	Combining materials - Describe what it means to combine materials - List methods and give examples of each to combine materials, namely - mixing and setting, e.g. Plaster of Paris and polyfilla, - Mixing and cooking, e.g. making dough, and porridge - Mixing and cooling, e.g. jelly powder and water to make jelly - Mixing, drying and firing, e.g. wet clay and straw to make bricks - List the properties of the new materials that are produced after combining the raw materials	The teacher demonstrates how to - mix Plaster of Paris and set the mixture in a mould – mixing and setting - make mealie meal pap – mixing and cooking - make jelly and set the jelly - mixing and cooling <u>Grade 5: First year</u> <u>Activity 1</u> List the equipment needed to perform an experiment to show how mixing and cooling produce processed materials. <u>Activity 2</u> Mix jelly in small groups of four learners and set the jelly in moulds and eat. <u>Activity 3</u> Copy a flow diagram of how to mix jelly. <u>Grade 5: Second year</u> <u>Activity 1</u>

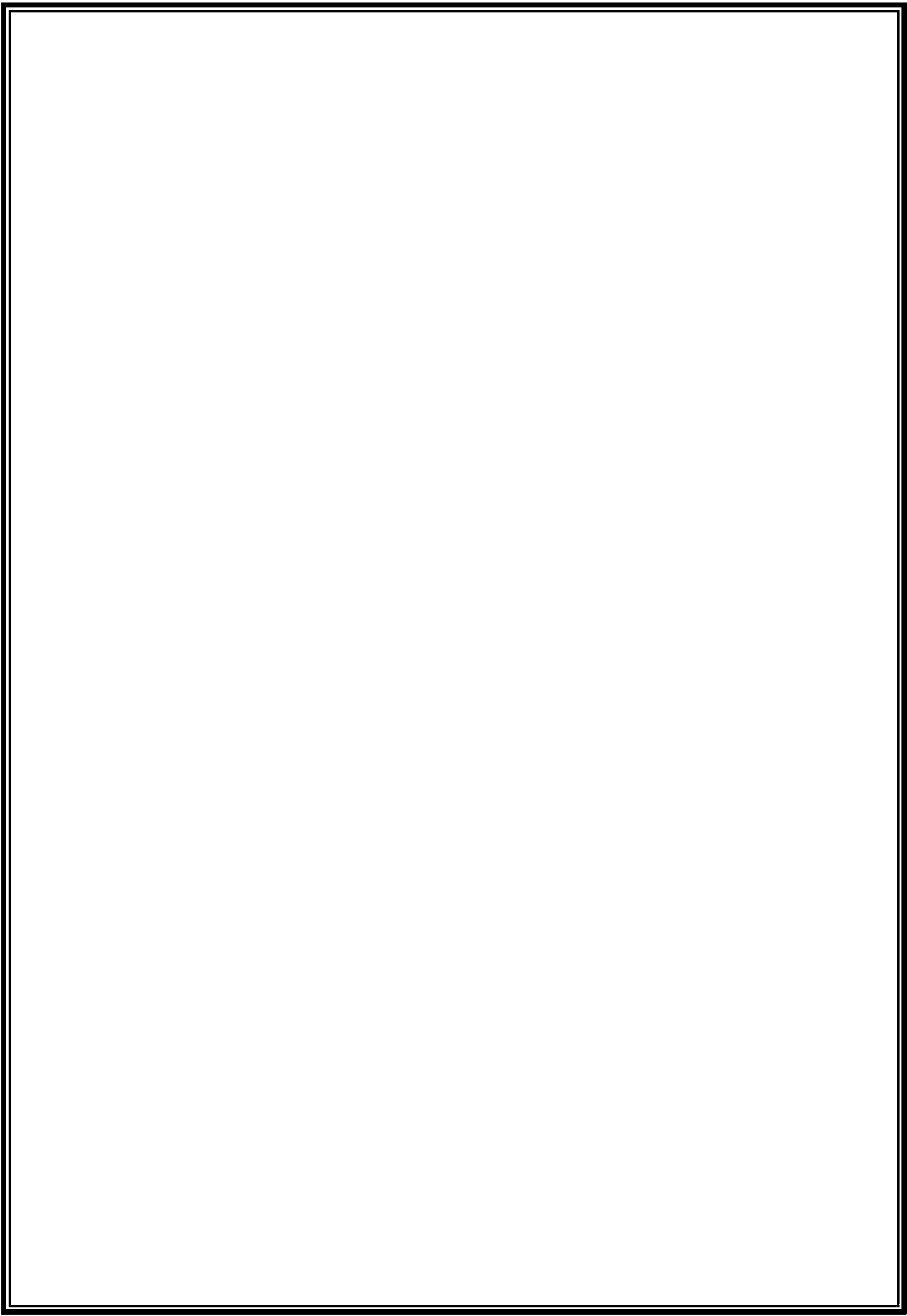
Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		Know and understand the terminology: mix, dissolve	List the equipment needed to perform an experiment to show how mixing and setting produce processed materials. <u>Activity 2</u> Mix Plaster of Paris in small groups of four learners and set the mixture in moulds. <u>Activity 3</u> Draw a flow diagram to illustrate how to mix Plaster of Paris. <u>Grade 5: Third year</u> <u>Activity 1</u> List the equipment needed to perform an experiment to show how mixing and cooking produce processed materials. <u>Activity 2</u> Prepare maize meal pap in small groups of two learners and enjoy the pap. <u>Activity 3</u> Draw a flow diagram to illustrate how to mix maize meal porridge.
7-8	Processed materials	- Know and understand the terminology: processed materials - Discuss the uses of Plaster of Paris, namely that Plaster of Paris is fire resistant and can be used for soft boards, to make casts and bandages, to fill cracks in walls and to produce art works - Explore different materials to see if they are durable and waterproof and can be	The teacher group examples of different materials together and demonstrate how to test the properties of different materials, namely water absorbency and strength. <u>Grade 5: First year</u> <u>Activity 1</u> Bring any two small items or pictures from home made with processed materials. Put all the items together that were brought to school and divide the items in groups according the raw materials that were used to produce the items. <u>Activity 2</u> Make a list of all the items in each group in your book.

Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		used for building • Explain the properties of ceramics and glass • Explain the uses of ceramics and glass namely to produce pottery items, tiles, basins, toilets and to use in construction glass is used as windows in buildings • Explain the properties of plastic and the uses of thereof, namely to produce plastic pipes, plastic coverings around electric wires and to produce dishes • Describe how traditional clay pots and bricks were produced • Distinguish between traditional methods of processing baskets, hat and mats, e.g. weaving, plating and coiling	<u>Activity 3</u> List the equipment needed to conduct an experiment to test Plaster of Paris for water absorbency and strength. <u>Activity 4</u> Test Plaster of Paris for water absorbency and strength. <u>Grade 5: Second year</u> The learners are grouped together in groups of four learners to execute the activities <u>Activity 1</u> Bring any two small items or pictures from home made with processed materials. Put all the items together that were brought to school and divide the items in groups according to the raw materials that were used to produce the items. <u>Activity 2</u> Make a list of all the items in each group in your book. <u>Activity 3</u> List the equipment needed to conduct an experiment to test concrete for water absorbency and strength. <u>Activity 4</u> Test concrete for water absorbency and strength and record your findings in a table. <u>Grade 5: Third year</u> The learners are grouped together in groups of two learners to execute the activities <u>Activity 1</u> Bring any two small items or pictures from home made with processed materials. Put all the

Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			items together that were brought to school and divide the items in groups according to the raw materials that were used to produce the items. <u>Activity 2</u> Make a list of all the items in each group in your book. <u>Activity 3</u> List the equipment needed to conduct an experiment to test ceramics for water absorbency and strength. <u>Activity 4</u> Test ceramics for water absorbency and strength and record your findings in a table.
9	Recycling	• Know and understand the terminology: water pollution, water borne diseases, water purification • Identify different types of water pollutants, namely litter and chemicals • Know that a clean supply of drinking water is important to people, plants and animals • List different processes to purify water, namely sieving, settling, decanting, boiling and adding chemicals • List places in your province that recycles water. • List the nearest buy-back and drop-off centres for recyclable items	Teacher demonstrates how to - make a water filter using a sieve, filter paper and funnel or a clean cloth or cotton handkerchief and elastic band - how to make a usable article using a can Learners are divided in small groups to do the following activities. <u>Grade 5: First year</u> <u>Activity 1</u> Purify dirty water by using the sieving method. <u>Activity 2</u> Practise to group and recycle recyclables at school. <u>Grade 5: Second year</u> <u>Activity 1</u> Purify dirty water by using the settling method. <u>Activity 2</u>

Grade 5 : Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		Visit the website of the National Recycling Forum in South Africa at www.recycling.co.za to find your nearest recycler in your area	Produce a usable article using empty cans collected in your community. <u>Activity 3</u> Practise to group and recycle recyclables at school. <u>Grade 5: Third year</u> <u>Activity 1</u> Purify dirty water by adding chemicals to the water. <u>Activity 2</u> Produce a decorative article using empty cans collected in your community. <u>Activity 3</u> Practise to group and recycle recyclables at school.





2.2. GRADE 5 TERM 3

Strand: Energy and change

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1 - 2	Energy in fuels	- Know and describe the terminology: fuel - Know and understand that fuel is a substance that can be eaten or burned to produce energy - Discuss how plants use the sun's energy to produce food - List fuels that humans use as energy sources, namely wood, coal, natural gas, candle wax, petrol and paraffin - Name the two prerequisites for fuels to keep burning, namely heat and oxygen - Identify and draw the fire triangle - Demonstrate how to stop a fire from burning, namely to cut off the oxygen supply - Describe how to extinguish an electrical fire in the home - Know the Learn not to Burn programme messages for young children, namely - Matches and lighters can burn you –	The teacher performs an experiment to show that oxygen is a prerequisite for fuels to keep burning. Three 10 cm candles are lit, the - first candle left uncovered to burn, - second candle covered with a small bottle and - third with a large bottle. The uncovered candle will burn the longest as oxygen is freely available and the candle covered with the small bottle will stop burning before the candle covered with the large bottle because of the limited amount of oxygen available in the small bottle. <u>Grade 5: First year</u> <u>Activity 1</u> Plan an experiment with two candles to show that fuel needs heat and oxygen to burn. <u>Activity 2</u> Sing the song from the Learn not to Burn programme for young children: I don't play with paraffin I don't touch I don't play It can burn me It can burn me So stay away Stay away

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		leave the alone - Stay away from paraffin – it can hurt you - Stay away from flames and fire – they can burn you - If your clothes catch on fire, stop, drop and roll	<u>Grade 5: Second year</u> <u>Activity 1</u> Perform an experiment with two candles to show that fuel needs heat and oxygen to burn <u>Activity 2</u> Make a list of possible electrical fire hazards in your classroom and tell how to extinguish an electrical fire. <u>Activity 3</u> Practise the Learn not to Burn message: “If your clothes catch on fire, stop, drop and roll” on command of the teacher. The learners move around in the class and immediately stop when the teacher command them to do so, then lie flat on the ground and cover their faces with their hands, and keep rolling over and over until the “flames” have been smothered. <u>Grade 5: Third year</u> <u>Activity 1</u> Make a list of possible electrical fire hazards in your home and explain how to extinguish an electrical fire. <u>Activity 2</u> Demonstrate the Learn not to Burn message: stop, drop and roll. <u>Activity 3</u> Use a fire extinguisher correctly.
3	Energy in fuels	• List ways to prevent fires namely: • Do not play with matches or lighters • Do not pour fuel onto a burning fire • Make sure candles are in sturdy candleholders and never put a	<u>Grade 5: First year</u> <u>Activity 1</u> List ways to prevent fires and demonstrate these to the class. <u>Grade 5: Second year</u>

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		candle near anything that might catch fire - Place fireguards in front of fireplaces - Do not dry clothes on or near a heater - Never use gas near drain pipes - Do not put your fingers in electrical appliances or wall plugs - Make sure to extinguish and outdoor campfire or braai - Know the Learn not to Burn messages for young children, namely: - Matches and lighters can burn you – leave them alone - If your house is on fire, get out and stay outside - Cool a burn with cool water - Fire fighters are helpers - Know how to get out of a building on fire	<u>Activity 1</u> Demonstrate the following Learn not to Burn message to the class: “If your house is on fire, get out and stay outside” to the class. Follow these steps: - Stay calm - Stay low on the ground to avoid the smoke - Hold onto one wall and crawl on your knees out of the building holding onto the wall - Never change holding onto the other wall as this may result in longer time to find the exit - Immediately call the fire brigade after exiting <u>Grade 5: Third year</u> <u>Activity 1</u> Draw an escape plan for your house and share your plan with the class. <u>Activity 2</u> Draw an escape plan for your class and choose a meeting place to meet after the escape has taken place. <u>Activity 3</u> Make a list showing the following emergency numbers: - Local fire brigade - Local ambulance - Emergency number to call from a cell phone, namely 112
4-5	Energy and electricity	Explain that energy can be stored in cells and batteries	The teacher demonstrates how to make a battery using two coins (one nickel and one copper), one lemon and 2 pieces of wire.

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		• Know that a battery is made of one or more cells • Explain that stored energy is controlled by a switch • Tell that the two ends of a battery is different and that one end is positive and the other negative • Describe an electrical circuit • Describe what happens when a circuit is broken • Discuss safety measurements when working with electricity • Discuss the national electricity grid • State the function of a national electricity grid • List the parts of the main electricity system • Describe how electricity gets to the appliances we use • List the sources of energy power stations utilize, namely coal, nuclear fuel, wind turbines, solar energy • List points how to use plugs and wall	<u>Grade 5: First year</u> <u>Activity 1</u> Practise how to put batteries in different appliances and take them out again. <u>Activity 2</u> Copy a diagram from the board of an electrical circuit in a torch into your book. <u>Activity 3</u> Repeat the teacher's experiment and make a lemon battery. <u>Grade 5: Second year</u> <u>Activity 1</u> Build your own electrical circuit using a battery, electrical wire and a light bulb. <u>Activity 2</u> Draw your electrical circuit to show how the battery, electrical wires and light bulb are connected. <u>Grade 5: Third year</u> <u>Activity 1</u> Practise how to wire an electrical plug. <u>Activity 2</u> Build your own electrical circuit with a switch, using a battery, electrical wire, a light bulb, two metal drawing pins, a metal paper clip and a plank about 8cmX 5 cm. <u>Activity 3</u> Make a list to indicate how you will use electrical appliances safely.

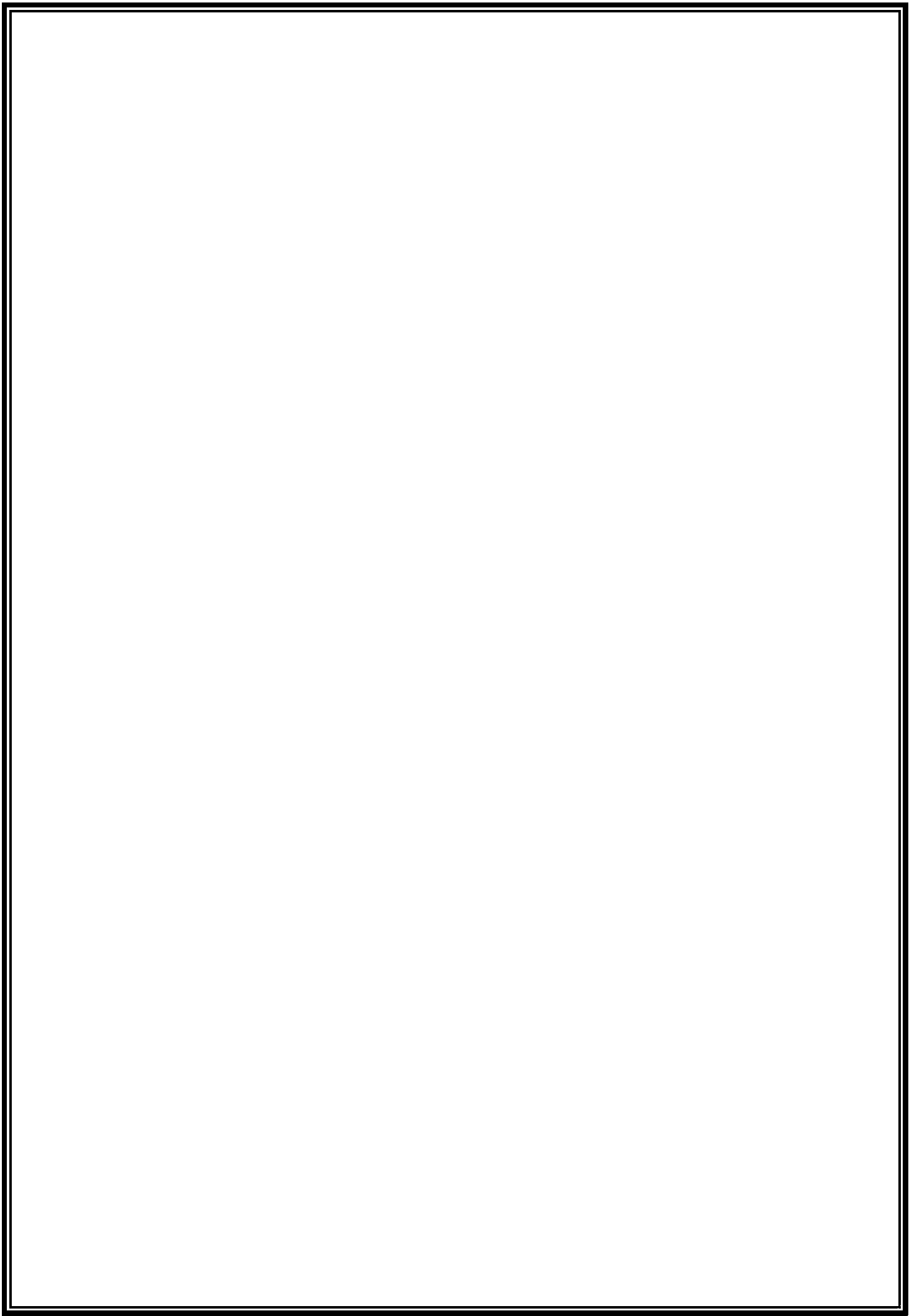
Grade 5 : Term 3

Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		sockets and electrical cords safely - List general safety precautions to take when using electricity and electrical appliances - Know and understand the following terminology: cells, positive end of cell and negative end of cell, stored energy, rechargeable cells, electrical circuit, electricity box, wall socket, plug, generator, pylon, and substation.	
6 - 7	Energy and movement	- Demonstrate how to make objects move by using stretched or twisted elastic and compressed springs - Explain that we store energy in an elastic band or spring when we twist or compress the band or spring - Demonstrate how to create movement energy by releasing an elastic band or compressed spring	The teacher demonstrates how - a hand-held catapult creates movement energy when the catapult is stretched and released. - the spring in a mousetrap creates movement energy upon releasing the spring - to make a wind-up toy using an elastic band, match stick and an empty cotton reel The teacher divides the learners in small groups of four learners to execute the activities. <u>Grade 5: First year</u> <u>Activity 1</u> Plan how to build a catapult. <u>Activity 2</u> Build your own catapult and demonstrate how to use the catapult safely. <u>Grade 5: Second year</u> <u>Activity 1</u>

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			Plan how to build a “bird in the cage” experiment to demonstrate how twisted elastic makes objects move. <u>Activity 2</u> Create a “bird in the cage” spinner using a rectangular shape card, a pen to draw the cage and the bird, a punch to punch holes on the left and right sides of the card, two elastic bands. <u>Grade 5: Third year</u> <u>Activity 1</u> Plan how to build a moving toy. <u>Activity 2</u> Build a moving toy.
8	Energy and movement	- Describe how a bicycle and a wheelbarrow moves - Know and understand the terminology: system, axle, turning axles, fixed axle - Identify different types of input energy that is suitable to make vehicles move, e.g. pram, bicycle, taxi, car	Teacher demonstrates how to build moving objects using wheels and axles. The teacher divides the learners in pair to complete the activities and provides examples with the equipment they need to complete the activity. <u>Grade 5: First year</u> <u>Activity 1</u> Build a toy vehicle with a turning axle. <u>Grade 5: Second year and second year</u> <u>Activity 1</u> Build a toy vehicle with a fixed axle. <u>Grade 5: Third year</u>

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			<u>Activity 1</u> Build a toy vehicle with a fixed as well as turning axle.
9	Recycling	- List examples of electronic waste - Discuss how electronic waste may be harmful to living and non-living things - Discuss pre-cautions to take when collecting electronic waste - Visit the electronic waste association website at www.ewasa.org for more information	<u>Grade 5: First year</u> <u>Activity 1</u> List the raw materials personal computers are made from and list the materials that may pose a risk to your health. <u>Activity 2</u> Practise to collect, group and recycle recyclables at school. <u>Activity 3</u> Discuss how you can re-use personal computers. <u>Grade 5: Second year</u> <u>Activity 1</u> List the raw materials cell phones are made from and list the materials that may pose a risk to your health. <u>Activity 2</u> Practise to collect, group and recycle recyclables at school. <u>Activity 3</u> Discuss how you can re-use cell phones. <u>Grade 5: Third year</u> <u>Activity 1</u> List the raw materials compact fluorescent lamps are made from and list the materials that may pose a risk to your health.

Grade 5 : Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			<u>Activity 2</u> Practise to group and recycle recyclables at school. <u>Activity 3</u> Discuss how you can re-use fluorescent lamps.



2.3. GRADE 5 TERM 4

Strand: Earth and beyond

Grade 5 : Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1 - 2	Surface of the earth	- State that the surface of the earth is made of rock and soil and is called the crust - List the determinants of life on earth, namely soil, air, water and sunlight - Distinguish between topsoil, subsoil and solid rock - Describe how topsoil is formed by rocks that are broken into tiny pieces - Distinguish between sandy soil, clayey soil and loamy soil - State that soil has air, water and very small living organisms in it - State and understand that dead plant and animals decompose in soil resulting in spreading nutrients through the soil to make it more fertile - Understand that soil forms very slowly in nature and cannot be replaced when lost - Know and understand the terminology:	The teacher executes an experiment - with soil in a bottle filled with water. Shake the bottle very well and then let the soil settle, the bubbles in the water show that spaces between the soil particles. After a while the water will form the top layer in the bottle, followed by the clayey layers, then sandy layer and gravel at the bottom. - with sandy, clayey and loamy soil to determine the amount of water let through each type of soil and the amount of water held by each type of soil. The teacher demonstrates to the learners how to record their findings in a table. The learners work together in pairs to complete the activities. <u>Grade 5: First year</u> <u>Activity 1</u> Locate and collect three different types of soil in the school gardens and repeat the teacher's experiment. Record your findings on the layers formed by different types of soil in your workbook. <u>Grade 5: Second year</u> <u>Activity 1</u> Select one type of soil provided by the teacher. Plant a seedling in the soil and measure the height of the seedling weekly for four weeks. Record your findings.

Grade 5 : Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		top soil, clayey soil, loamy soil, decompose, erosion	<u>Grade 5: Third year</u> <u>Activity 1</u> Select one type of soil provided by the teacher. Plant a seedling in the soil and measure the height of the seedling weekly for four weeks. Record your findings. Compare the height of the seedlings in the different types of soil. Give three reasons why you think the growth of the seedlings varies. Provide reasons why different types of soil influence plant growth.
3	Sedimentary rocks	• Know and understand the terminology: sediments, sedimentary rock, compact, shale, sandstone, limestone • Describe what sedimentary rock means, namely rock formed over a long time and consists of different layers of rock • Describe how sedimentary rock is formed, namely that mud, dust and sand are deposited in low-lying areas, the layers become compacted and hardens to form sedimentary rock • List a characteristic of sedimentary rock, namely that layers are visible within the rock • List examples of sedimentary rock, namely shale, sandstone and limestone • Give examples of the uses of limestone,	<u>Grade 5: First year</u> <u>Activity 1</u> Make a sedimentary layer model. <u>Grade 5: Second year</u> <u>Activity 1</u> Make a sample of sedimentary rock from rocks (gravel), sand and water. <u>Grade 5: Third year</u> <u>Activity 1</u> Make sedimentary rock with rocks (gravel), sand and clay.

Grade 5 : Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		shale and sandstone	
4-5	Fossils	- Know and understand the terminology: fossils, preserved - Tell how fossils are formed - Tell why fossils are important to humans - Name one fossil site in your province - Name a region in South Africa where dinosaurs lived many years ago - Name the region where human fossils have been found in South Africa - Read and understand a time chart showing when which living things existed	The teacher demonstrates how to make a plant fossil, using plasticine, plaster of Paris and a leaf with distinct veins. Roll the plasticine out about 10 mm thick, press the leaf into the plasticine, build the outer edges of the plasticine up, peel the leaf off, fill the plasticine mould with a mixture of plaster of Paris and leave to set. When the plaster of Paris has set, peel off the plasticine and you will have a leave “fossil”. A mould can also be made using damp, fine sand similar to sea sand. <u>Grade 5: First year</u> <u>Activity 1</u> Plan how to make a leave “fossil”. <u>Activity 2</u> Make your own leave “fossil”. <u>Grade 5: Second year</u> <u>Activity 1</u> Plan how to make a sea shell “fossil”. <u>Activity 2</u> Make your own sea shell “fossil”. <u>Grade 5: Third year</u> <u>Activity 1</u> Plan how to make a sea animal “fossil”. <u>Activity 2</u>

Grade 5 : Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			Make your own sea animal “fossil”, e.g. crab sea star.
6 - 8	Movement of the earth and planets	- Know that the movement of the earth and planets results in different seasons - Name the four seasons and the months of the year during which we experience the different seasons - Identify the weather symbols for sunny, cloudy, partly cloudy and rainy weather - Identify the weather symbols in different media - Explain that we measure temperature with a thermometer - Read and record temperatures on a thermometer correctly - Explain how to measure the direction of wind - Read a rain gauge correctly and record rainfall correctly	The teacher demonstrates how to make a - wind-vane to show the direction the wind is coming from - rain gauge with a jar and a strip of masking tape The teacher divides the learners in small groups of four learners to execute the activities. <u>Grade 5: First year</u> <u>Activity 1</u> Cut a weather forecast chart from a newspaper and discuss the predicted weather according to the symbols on the weather chart. <u>Activity 2</u> Draw a table with two rows and seven columns. Write the names of the days in the top row and record the daily temperature over one week. <u>Activity 3</u> Plan how to make a wind-vane. <u>Activity 4</u> Make your own wind-vane and blow in different directions to establish the direction of the “wind” <u>Grade 5: Second year</u> <u>Activity 1</u> Cut and paste weather symbols in your workbook. List the words that represent the different weather symbols. <u>Activity 2</u>

Grade 5 : Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			Draw a table with three rows and seven columns. Write the names of the days in the top row and record the daily temperature and wind direction over one week. <u>Activity 3</u> Plan how to make a rain gauge. <u>Activity 4</u> Make your own rain gauge to measure rainfall and record the rainfall. <u>Grade 5: Third year</u> <u>Activity 1</u> Explain the weather forecast chart for the previous day. <u>Activity 2</u> Examine the weather forecast chart and record the town with the highest and lowest temperatures and wind speed. <u>Activity 3</u> Plan how to make a model to deliver weather forecasts. <u>Activity 4</u> Make a model to deliver weather forecasts.
9	Recycling	- Discuss what happens to rubbish that we don't recycle - Locate the nearest landfill to your school and home - Practise to recycle different recyclables at your school	<u>Grade 5: First year</u> Practise to collect, group and recycle recyclables at school. <u>Grade 5: Second year</u> Practise to collect, group and recycle recyclables at school. <u>Grade 5: Second year</u> Practise to collect, group and recycle recyclables at school.

